	SCHOOL OF COMPUTER ENGINEERING **III SEMESTER B.TECH.(IT/CCE/CSE/AIML/DSE)** **Data Communication and Computer Networks (CSS 2102)** **Midterm Examination-SCHEME** Exam Date : 13-9-25
Q.No	Question
1	A 1000-byte frame is sent using Stop-and-Wait ARQ. The datarate is **1 Mbps**, and one-way propagation delay is **25 ms**. What is the channel utilization ? 1. **1.9 % 2. 0.19% 3. 19% 4. 3.8%
2	In a communication system using Selective Repeat ARQ, the sequence number field is 8 bits long. To avoid ambiguity between old and new frames with the same sequence number, what is the maximum allowable sender window size? 1. **128 2. 255 3. 256 4. 127
3	If an organization has 10 nodes, and wants to link the nodes using mesh topology, the number of links required for duplex communication is: a. 10 b. ** 45 c. 90 d. 20
4	Hop to Hop delivery is the responsibility of a. Network Layer b.Physical layer c. Transposrt Layer d. ** Data link Layer
5	Find the number of addresses in the range if the first address is 152.79.52.30 and the last address is 152.79.56.78 1. 52 2. 53 3. **1073 4. 1072

6	What is the result of performing a bitwise AND operation between the IP address 29.135.14.56 and the subnet mask 255.255.140.0 1. **29.135.12.0 2. 29.135.14.0 3. 29.135.12.56 4. 29.135.14.56
7	Find the class of the following IPv4 address: 11000010 10011011 10101001 11111111 a. Class A b. Class B c. **Class C d. Class D
8	In the classful addressing if the network is divided into 8 subnetworks and if one of the addresses in one of the subnets is 141.14.120.77, then the subnet address is: a. 141.14.255.255 b. **141.14.96.00 c. 141.14.0.0 d. 141.14.120.0
9	Which of the following does not have a Net ID and Host ID? 1. Class A 2. Class B 3. Class C 4.** Class D
10	If you put all binary 1's in host identifier in an IP address, you get the _____ 1. ** broadcast address 2. first valid IP address 3. last valid IP address 4. network address
11	In classful addressing, an organization is granted the block 16.0.0.0/8. The administrator wants to create 8 fixed-length subnets. Analyze the network and determinea. The subnet mask. b. The number of addresses in each subnet. c. The first and the last address in the first subnet. d. The first and the last address in the last subnet (subnet 500). a.Subnet Mask **[1M]** Since netid=8 ,first address of the entire network is 16.0.0.0/8 and last address is 16.255.255.255/8 Entire network is divided into 8 subnets = 2^3 Subnet Length = = 8 +3 =11 So subnet mask is 255.224.0.0 (/11) b. Number of addresses per subnet is $2^{21=}$ 20,97,152 **[1M]**

	c. First Subnet Network: 16.0.0.0 [**1M]** First Host: 16.0.0.1 Last Host: 16.31.255.254 Broadcast: 16.31.255.255 d. Last Subnet Network: 16.224.0.0 [**1M**] First Host: 16.224.0.1 Last Host: 16.255.255.254 Broadcast: 16.255.255.255
12	A sender needs to transmit a total of 800,000 bits of data using the Stop-and-Wait ARQ protocol, where each frame carries 1000 bits. The transmission can take place over two possible physical links. Link A has a one-way distance of 400 km, a propagation speed of 2×10^8 m/s, and a transmission time per frame of 1 millisecond. Link B has a one-way distance of 100 km, the same propagation speed of 2×10^8 m/s, but a transmission time per frame of 2 milliseconds. For both links, assume that acknowledgment transmission time and processing delays are negligible, and that no frames are lost or corrupted. (a) Calculate the total time required to send all the data over each link. (b) Based on your results, analyze which link is faster and explain the tradeoff observed between propagation delay and transmission time. **Solution** The sender transmits 800 frames of 1000 bits each. Nr of frame = 800,000/1000=800 For Stop-and-Wait, cycle time = transmission time + 2 × propagation delay. Link A: propagation delay =Distance/Speed= 2 ms, transmission = 1 ms → cycle = 2 x 2 + 1=5 ms. . Total time = 800 × 5 ms = 4.0 s. 1 M Link B: propagation delay = 0.5 ms, transmission = 2 ms → cycle = 3 ms. Total time = 800 × 3 ms = 2.4 s. 1 M Link B is faster because its shorter propagation delay outweighs the longer transmission time. 1 M

13	A communication engineer is tasked with designing a digital modulation scheme for a telemetry system. The available frequency band is from 200 kHz to 400 kHz. To efficiently transmit data, the system must encode 3 bits per symbol using Multiple Frequency Shift Keying (MFSK). As the engineer, determine **a)** How many distinct frequency levels should be used. **b)** The symbol (baud) rate of the system. **c)** The approximate bandwidth requirement for this design. **Answer** **c)** W_d =Bandwidth available = 400kHz -200KHz = 200KHz **1 Mark** **a)** Number of bits Per Symbol = L= 3 **1 Mark** Number of Frequency levels= $M=2^3=8$ $W_d= M(2*f_d)$ f_d = 200/(2*8) = 12.5 kHz f_c = 300 kHz 8 Different frequencies- $f_i=f_c +(2i-1- M)f_d$ 1st frequency f_1 =300+(2-1-8) 12.5 = 300 -87.5 =212.5 kHz f_2 =300+(2*2-1-8) 12.5 = 300 -62.5 =237.5 kHz f_3 =300+(2*3-1-8) 12.5 = 300 -37.5 =262.5 kHz f_4 =300+(2*4-1-8) 12.5 = 300 -12.5 =287.5 kHz f_5 =300+(2*5-1-8) 12.5 = 300 +12.5 =312.5 kHz f_6 =300+(2*6-1-8) 12.5 = 300 + 37.5 =337.5 kHz f_7 =300+(2*7-1-8) 12.5 = 300 +62.5 =362.5 kHz f_8=300+(2*8-1-8) 12.5 = 300 +87.5 =387.5 kHz **b)** Baud rate = Data rate / Number of bits per symbol = **6 Mbps/3 = 3 Mega baud** **1 Mark**
14	Eight voice channels, each occupying a bandwidth of 3.2 kHz, are to be transmitted over a link of 25.6 kHz available in the range 140.0 to 165.6 kHz. The channels are first combined into two TDM frames, each carrying four voice channels, and then the two frames are multiplexed into the link using FDM without guard bands. Show, using the frequency domain, how the sub-bands are allocated and discuss how TDM and FDM together allow all eight channels to be transmitted simultaneously. Ans: Each voice channel = 3.2 kHz. One TDM frame (4 channels) needs 3.2×4= (32×4)/10=128/10=12.8 kHz.

	Two frames need 2×12.8=25.6 Available link width = 25.6 kHz **[1M]** Each sub-band width = 12.8 kHz. Place them contiguously: • Lower sub-band: 140.0 to 140.0+12.8=152.8 kHz. f_{cA} = (140.0+152.8)/2=292.8/2=146.4 kHz. • Upper sub-band: 152.8 to 152.8+12.8=165.6 kHz. f_{cB} = (152.8+165.6)/2=318.4/2=159.2 kHz. **[1M]** Within each sub-band a TDM frame time-multiplexes four voice signals into four time slots; then FDM places the entire TDM frame into its allocated frequency sub-band (one frame in 140.0–152.8 kHz, the other in 152.8–165.6 kHz). Thus all eight channels travel simultaneously **[1M]**
15	Apply the CRC method to the following data: Message D = 1101101010 and Generator Polynomial P = 101011 (i) Determine the transmitted message. (ii) Verify correctness of the received message assuming no transmission error.

CRC Computation (Transmitter & Receiver Side)

Transmitter Side Computation [1.5 Marks]

D=1101101010 (10 bits)
P=101011 (6 bits → degree = 5)
Append 5 zeros → Dividend = 1101101010 00000 = 110110101000000
We divide 110110101000000 by 101011 (modulo-2 / XOR division). Each row below shows:

1. the current 6-bit window used for XOR,
2. XOR with divisor 101011,
3. the XOR result,
4. the next 6-bit window formed after bringing down the next bit (if any).

Step 1:
window = 110110
XOR 101011
result 011101
bring down next bit -> next window = 111011

Step 2:
window = 111011
XOR 101011
result 010000
bring down next bit -> next window = 100000

Step 3:
window = 100000
XOR 101011
result 001011
bring down next bit -> next window = 010111

Receiver Side Verification [1.5 Marks]

Received frame = 110110101001100
Divisor = 101011

Step 1:
window = 110110
XOR 101011
result 011101
bring down next bit -> next window = 111011

Step 2:
window = 111011
XOR 101011
result 010000
bring down next bit -> next window = 100000

Step 3:
window = 100000
XOR 101011
result 001011
bring down next bit -> next window = 010111

Step 4:
window = 010111
(leading 0 → skip XOR)
result 010111
bring down next bit -> next window = 101110

Step 5:
window = 101110
XOR 101011
result 000101
bring down next bit -> next window = 001010

Step 6:
window = 001010
(leading 0 → skip XOR)
result 001010
bring down next bit -> next window = 010101

Step 7:
window = 010101
(leading 0 → skip XOR)
result 010101

Step 4:
window = 010111
(leading bit 0 → effectively no XOR with divisor; window stays 010111)
result 010111
bring down next bit -> next window = 101110

Step 5:
window = 101110
XOR 101011
result 000101
bring down next bit -> next window = 001010

Step 6:
window = 001010
(leading 0 → no XOR)
result 001010
bring down next bit -> next window = 010100

Step 7:
window = 010100
(leading 0 → no XOR)
result 010100
bring down next bit -> next window = 101000

Step 8:
window = 101000
XOR 101011
result 000011
bring down next bit -> next window = 000110

Step 9:
window = 000110
(leading 0 → no XOR)
result 000110
bring down next bit -> next window = 001100

Step 10:
window = 001100
(leading 0 → no XOR)
result 001100
no more bits to bring down → remainder = last 5 bits = 01100

Step 3: Append remainder to data

bring down next bit -> next window = 101011

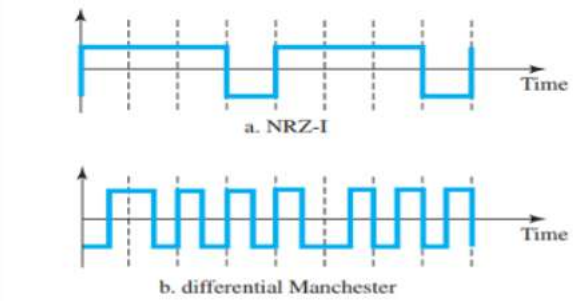
Step 8:
window = 101011
XOR 101011
result 000000
bring down next bit -> next window = 000000

Step 9:
window = 000000
(leading 0 → skip XOR)
result 000000
bring down next bit -> next window = 000000

Step 10:
window = 000000
(leading 0 → skip XOR)
result 000000
no more bits to bring down → remainder = 00000

Final Check: Remainder = 00000 → No error detected ✅

	Transmitted frame = 1101101010 01100	
16	Compare circuit switching and packet switching in terms of setup and resource usage. Further, calculate the total transmission delay when a 2000-bit packet traverses 3 store-and-forward switches connected with 1 Mbps links. Ans: Compare setup & resource usage → *1 mark.* Transmission delay Per link = 2000/1,000,000=0.002 s=2 ms → *1 mark.* 4 links (source + 3 switches + destination) = 4×2=8 ms → *1 mark.*	
17	A 1000-byte packet is sent from host A to host B through two routers (R1 and R2) over three store-and-forward links, each operating at 1 Gbps and having a propagation delay of 2 ms. Ignoring processing delays, calculate the total time taken for the last bit of the packet to reach host B Solution: With processing delay ignored, the 1000-byte packet (8000 bits) needs $8000/10^9=8\,\mu s$ transmission per 1 Gbps link, so across three store-and-forward links that's $3\times8=24\,\mu s$; 1M Each link adds 2 ms propagation delay, totalling $3\times2=6\,ms$; Therefore, the last bit reaches B after $6\,ms + 24\,\mu s = 6.024\,ms$. 1 M	
18	Find the class of the following IP address: also find first and last address **(a)15.25.14.100** (b)**130.45.67.89** Ans : a. class A: First address 15.0.0.0 Last address 15.255.255.255 1M b. class B: First address 15.0.0.0 Last address 15.255.255.255 1M	

19	Find the 8-bit data stream for given figure.  Ans: a. NRZ-I: 10011001. b. Differential Manchester: 11000100.